

Index laws



1

$a \quad 2^{21}$

$e \quad 5^{48}$

$i \quad 19^5$

$b \quad 22^8$

$f \quad 15^4$

$j \quad 12^7$

$c \quad 11^4$

$g \quad 23^{79}$

$k \quad 13^8$

$d \quad 7^5$

$h \quad 17^8$

$l \quad 15^{27}$

2

$a \quad 8$

$e \quad 3$

$i \quad 5$

$b \quad 7$

$f \quad 6$

$j \quad 2$

$c \quad 34$

$g \quad 5$

$d \quad 5$

$h \quad 38$

3

$a \quad 1\,679\,616$

$e \quad 16\,807$

$i \quad 36$

$b \quad 9\,261$

$f \quad 4\,096$

$j \quad 1$

$c \quad 1\,024$

$g \quad 121$

$k \quad 1\,728$

$d \quad 390\,625$

$h \quad 729$

$l \quad 36$

Negative bases

4

$a \quad (-11)^{13}$

$e \quad -12$

$i \quad (-7)^5$

$b \quad (-21)^8$

$f \quad (-30)^3$

$j \quad (-6)^2$

$c \quad -15^2$

$g \quad 8^3$

$d \quad (-3)^7$

$h \quad 11^{11}$

5

$a \quad -7$

$e \quad 49$

$i \quad 3$

$b \quad -3$

$f \quad 12$

$j \quad -55$

$c \quad 50$

$g \quad -2$

$d \quad 2$

$h \quad -21$

6

$a \quad 256$

$e \quad -27$

$i \quad 1\,225$

$b \quad 64$

$f \quad -3\,125$

$j \quad 531\,441$

$c \quad -243$

$g \quad -216$

$k \quad 64$

$d \quad -8\,000$

$h \quad -1\,977\,326\,743$

$l \quad -8\,000$



Fractional bases

7

a $\frac{1}{3^4}$

b $\frac{3^3}{8^3}$

c $\frac{1^8}{4^8}$

d $\frac{5^2}{2^2}$

e $\frac{10^5}{33^5}$

f $\frac{2^6}{35^6}$

g $\frac{5^3}{18^3}$

h $\frac{29^7}{41^7}$

i $\frac{11^9}{13^9}$

j $\frac{20^2}{3^2}$

k $\frac{17^4}{4^4}$

l $\frac{31^5}{50^5}$

8

a 3

b 3

c 2

d 1

e 3

f 16

g 9

h 625

i 2

j 121

k 5

l 5

9

a $\frac{1}{9}$

b $\frac{27}{125}$

c $\frac{1}{32}$

d $\frac{49}{121}$

e $\frac{27}{512}$

f $\frac{144}{169}$

g $\frac{125}{729}$

h $\frac{16}{81}$

i $\frac{4\,096}{15\,625}$

j $\frac{8}{125}$

k $\frac{25}{49}$

l $\frac{1\,000}{2\,744}$



10

a No

b Yes

c No

d Yes

Negative indices



11 Less than 1.

12

a $\frac{1}{a^9}$

b a^n

c $\frac{1}{4a^9}$

d $\frac{b^m}{a^n}$

e $\frac{1}{p^2}$

f $\frac{3}{p^4}$

g $\frac{7}{x^9}$

h $\frac{q^3}{p^2}$

i $\frac{8}{p^3}$

j $\frac{2y^3}{x^8}$

k $\frac{c^3}{6b^7}$

l $\frac{p^4}{m^5n^4}$

m $\frac{3a^4}{b^5c^7}$

n $\frac{l^7}{16k^3}$

o $\frac{g^2}{f^7h^8}$

p $\frac{s^4u^6}{5t^9}$

13

a u^{-4}

b $2r^{-6}$

c xy^{-2}

d $4x^{-6}a^{-7}$



14

a $20y^6$

b $28a^2$

c $\frac{-15}{x^4}$

d $\frac{12}{y^5}$

e $8h^7$

f $\frac{6}{y^7}$

g $\frac{16}{y^3}$

h $25m^3$

i $\frac{3}{x^7}$

j $\frac{3}{x^4}$

k $3x^8$

l $\frac{3}{2c^2}$

m $\frac{5h^9}{4}$

n $3x^8$

o $\frac{3}{2p}$

p $\frac{9}{4w^{10}}$

15

a $\frac{1}{8m^3}$

b $\frac{256}{m^{24}}$

c $\frac{p^8}{9}$

d $\frac{1}{9y^4}$

e $\frac{64}{y^3}$

f $\frac{y^4}{x^5}$

g $\frac{y^{36}}{x^{28}}$

h $5x^5$

i $\frac{5}{x^4}$

j $\frac{81}{z^4}$

k $\frac{q^7}{p^3}$

l $\frac{x^8}{y^{16}}$

16

a $n = -2$

b $n = -3$

c $n = -4$

d $n = -3$

17 $(x^4y^{-3})^{-4} = \frac{y^{12}}{x^{16}}$

18 The -1 index only applies to the term directly in front of it. In this case it only applies to the a , not the 5. The correct working is $5a^{-1} = \frac{5}{a}$.

19

a $32y^8$

b $60y^5$

20

a $\frac{10z^9}{21x^3y^7}$

b $\frac{160y^6}{27x}$